

From Click to Checkout

E-commerce Customer Journey Analysis for Maven Toy Store



Siddhi Kataria

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Data Analyst

1. Project Overview

Every customer purchase begins with a single click. However, not every visitor who lands on an e-commerce website completes a purchase. Understanding how customers navigate the website—from their first visit to the final checkout—is essential for improving conversions and maximizing revenue.

This project analyzes Fuzzy Factory's e-commerce data using **Python, MySQL, and Power BI** to identify customer behavior, evaluate website performance, measure sales trends, and uncover opportunities to improve the online shopping experience.

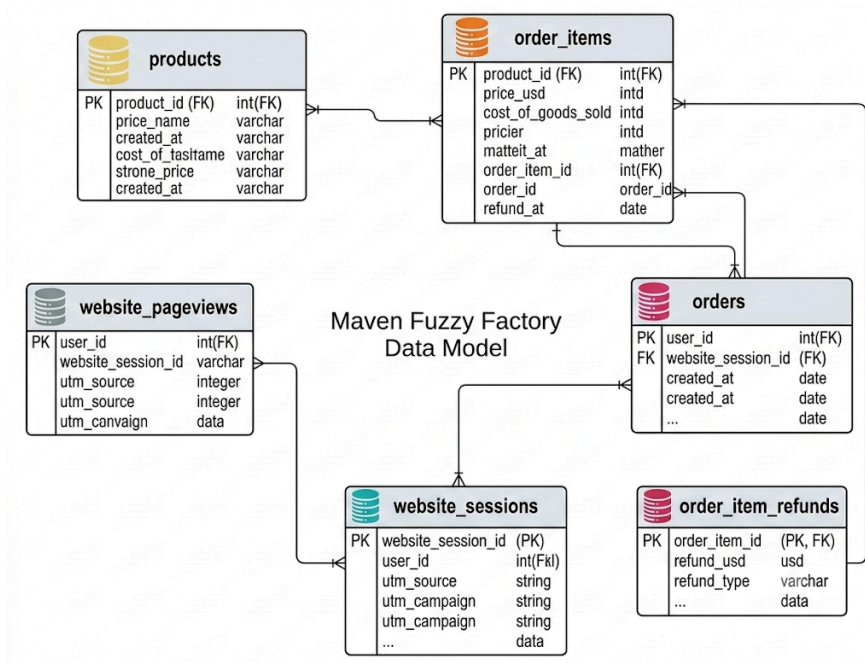
The analysis follows the customer's journey from **website visit** → **product discovery** → **purchase** → **post-purchase**, providing actionable business recommendations at each stage.

2. Business Objectives

- Analyze customer acquisition channels.
- Evaluate website engagement and user behavior.
- Assess product performance and purchase trends.
- Identify opportunities to improve conversion.
- Provide data-driven business recommendations.

3. Data Preparation

The raw datasets were imported into **Python (Pandas)**, where missing values, duplicate records, inconsistent formats, and data types were cleaned and standardized. The cleaned data was then loaded into a **MySQL** relational database, where primary and foreign key relationships were established to maintain data integrity. Our dataset in a nutshell can be seen in the diagram below.



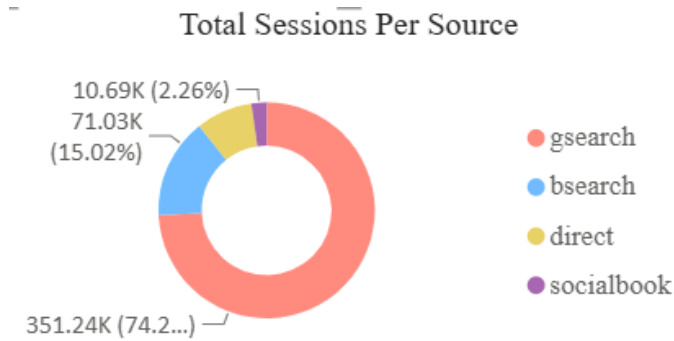
The MySQL database was connected to **Power BI**, enabling the creation of interactive dashboards to analyze website traffic, customer behavior, product performance, sales trends, and conversion metrics.

4. The Customer Journey

1. Acquiring Customers – Where Does the Journey Begin?

Every customer starts by visiting the website through a marketing channel. The first step was to evaluate which traffic sources generated not only visitors but actual customers. While Google Search drives higher overall traffic and revenue, Bing Search achieves a significantly better conversion rate.

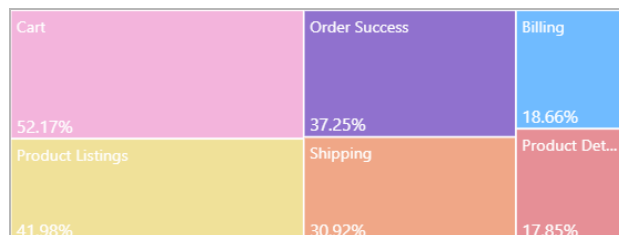
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utm_source	revenue	sessions	orders	conversion rate
bsearch	313.48k	71032	5262	7.41
direct	174.62k	39917	2856	7.15
gsearch	1428.15k	351237	23852	6.79
socialbook	22.26k	10685	343	3.21

2. Turning Visitors into Customers

After arriving on the website, the next objective is understanding how many visitors actually complete a purchase.

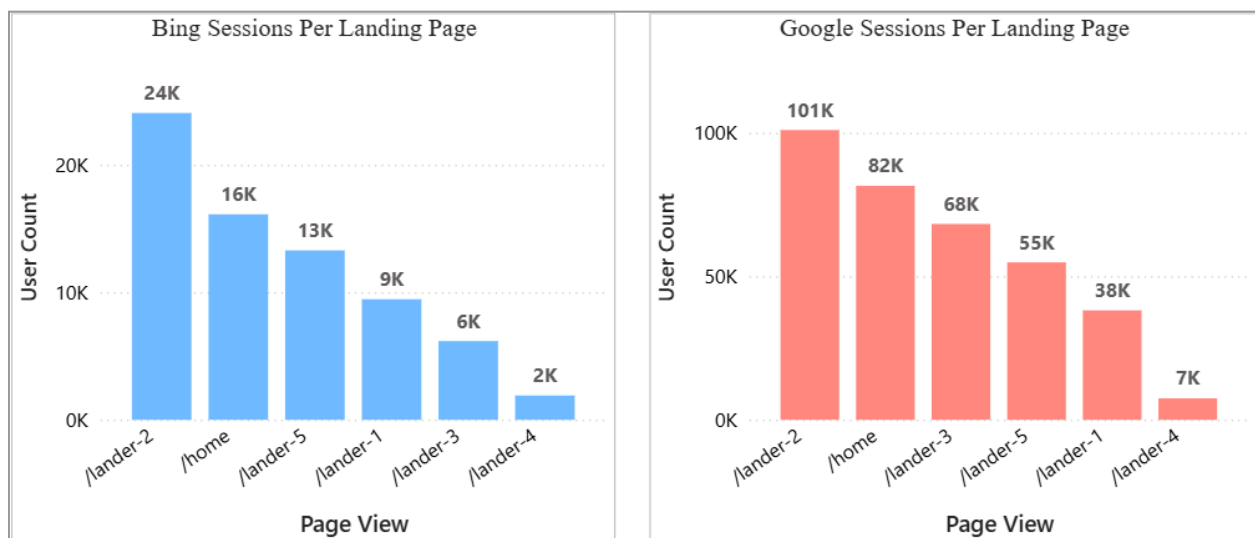


session type	total sessions	session %
No Purchase	440558	93.17
Purchased	32313	6.83

Although the website attracts substantial traffic, only **6% of sessions result in a completed purchase**, meaning approximately **93% of visitors leave without buying**. Furthermore, more than **half of all visitors abandon their journey before adding products to the cart**, indicating that the largest customer loss occurs early in the purchasing process.

3. The First Impression – Landing Page Performance

Landing pages create the customer's first impression and often determine whether visitors continue exploring or leave the website.



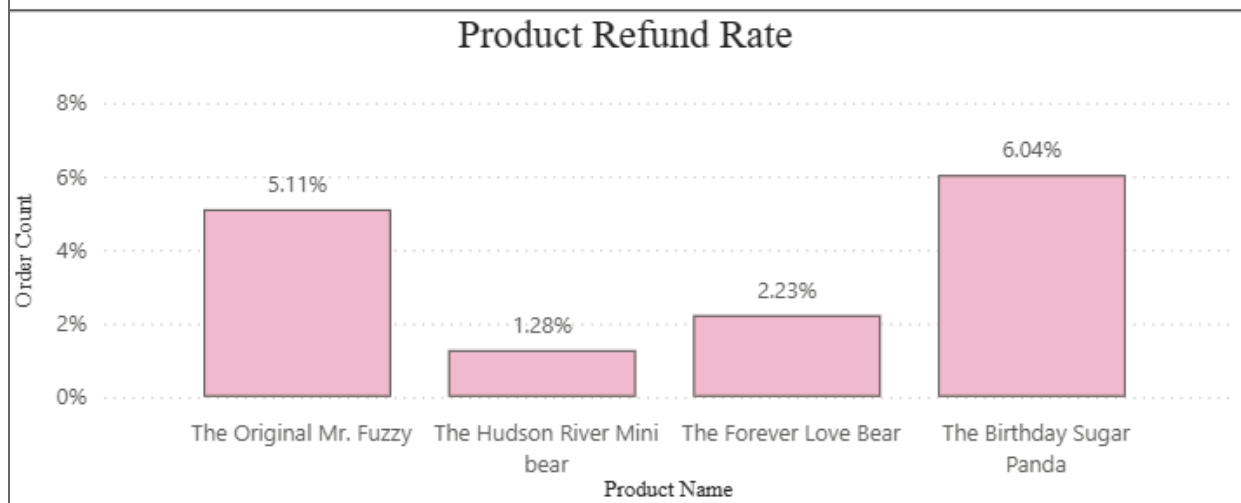
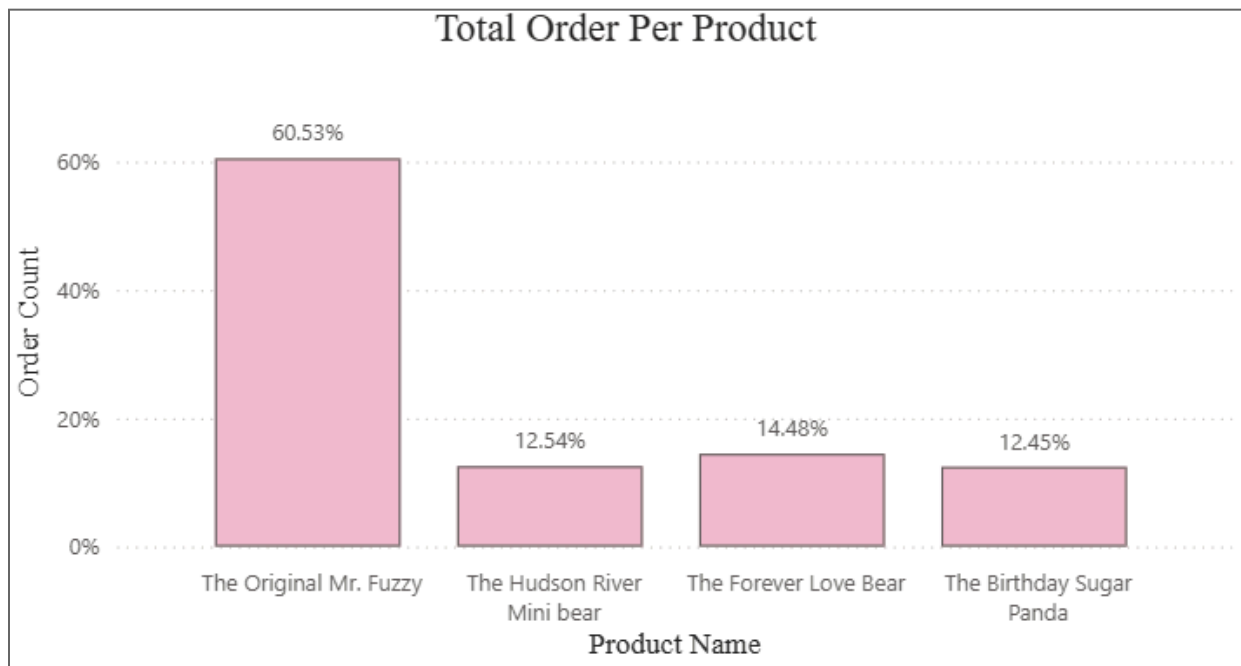
pageview_url	total sessions	total orders	product_conversion_rate
/lander-5	68166	6930	10.17
/lander-2	131170	10128	7.72
/lander-4	9385	708	7.54
/home	137576	9711	7.06
/lander-1	47574	2157	4.53
/lander-3	79000	2679	3.39

The analysis shows that **/lander-5** achieves the highest conversion rate, yet the majority of visitors arriving through both Google Search and Bing Search are directed to **/lander-2**, followed by the homepage.

Redirecting more traffic toward the highest-performing landing page presents an immediate opportunity to improve website conversion without increasing marketing expenditure.

4. Product Discovery – Are Customers Finding the Right Products?

After entering the website, customers begin exploring products before making a purchase decision.



pageview_url	total page views	total orders	product_conversion_rate
/the-birthday-sugar-panda	19046	3068	16.11

/the-forever-love-bear	26033	4803	18.45
/the-hudson-river-mini-bear	2610	581	22.26
/the-original-mr-fuzzy	162525	23861	14.68

The **Original Mr. Fuzzy** generates approximately **60% of total sales**, making it the company's strongest revenue contributor. However, despite its popularity, it records the lowest conversion rate among the product range.

In contrast, the **Hudson River Mini Bear** achieves the highest conversion rate while maintaining the lowest refund rate, suggesting that it is a high-performing product that currently lacks sufficient visibility.

Meanwhile, the **Birthday Sugar Panda** experiences the highest refund rate, indicating potential product quality or customer expectation issues that directly impact profitability.

5. Building Customer Loyalty

Acquiring customers is only one part of sustainable growth. Long-term profitability depends on encouraging customers to return.

The analysis indicates that **98% of customers never return to make another purchase**, revealing a significant opportunity to improve post-purchase engagement and customer retention strategies.

customer_type	customers
One Time Customer	31105
Repeat Customer	591

5. Key Business Insight

The customer journey analysis identified several critical opportunities to improve business performance:

- Bing Search generates fewer visitors than Google Search but delivers significantly higher-quality traffic.
- Most customers abandon the purchasing journey before reaching the checkout process.
- The highest-converting landing page is currently underutilised.
- High-performing products are not receiving sufficient customer exposure.
- Product refund rates vary substantially across the catalogue, reducing overall profitability.
- Customer retention remains extremely low, limiting long-term revenue growth.

6. Strategic Recommendations

Based on the analysis, the following business initiatives are recommended:

- **Optimise Marketing Investment** by shifting more advertising budget toward higher-converting acquisition channels such as Bing Search and Direct Traffic

while reducing spend on lower-performing channels.

- **Improve Landing Page Strategy** by directing more traffic to the highest-converting landing page to maximise visitor-to-customer conversion.
- **Enhance Product Merchandising** by increasing visibility for high-converting products such as the Hudson River Mini Bear while reducing dependence on a single flagship product.
- **Reduce Checkout Friction** by simplifying the customer journey and addressing major abandonment points between product selection, cart, and billing.
- **Reduce Refund Costs** by investigating quality concerns associated with the Birthday Sugar Panda and improving customer expectations through better product information.
- **Strengthen Customer Retention** by implementing post-purchase email campaigns, loyalty programmes, and personalised product recommendations to encourage repeat purchases.

7. Business Impact

This analysis demonstrates how combining customer journey analytics with business intelligence can help organisations make evidence-based decisions that improve marketing effectiveness, website usability, product performance, and customer retention.

Rather than focusing solely on increasing website traffic, the project identifies where customers disengage throughout the purchasing process and recommends targeted improvements that can increase conversion rates, maximise revenue, and support sustainable business growth.